Vision-Language Models for Soccer Video Retrieval: Moment Retrieval and Action Spotting via Sliding-Window Embedding Search

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Abstract

We study natural language grounding in broadcast soccer video across two tasks: moment retrieval, where a free-form caption must be localized to a single timestamp, and action spotting, where a class label must be matched to all its occurrences. Both are cast as cross-video problems, requiring search across 471 games and roughly 1.1 million sliding-window candidates. We propose a unified pipeline that extracts per-frame embeddings offline and ranks windows by cosine similarity to the text query at inference, requiring no task-specific architecture. We instantiate it with CLIP ViT-L/14 and Qwen3-VL-Embedding-8B as drop-in backbones, and introduce an oracle diagnostic framework that decomposes retrieval error into game-selection and within-game components. Zero-shot CLIP achieves 0.77% R@1 on moment retrieval and 1.83% mAP@5s on action spotting, with the oracle analysis showing a 5.7× gap attributable to game-selection failure rather than event localization. We further propose a pairwise fine-tuning recipe for CLIP that operates on a single V100 without large-batch contrastive requirements.

1 Introduction

A broadcast soccer match runs over 90 minutes and contains dozens of semantically distinct events—goals, fouls, corners, substitutions—embedded in long stretches of visually similar play. We study two natural language grounding tasks over this footage: **moment retrieval**, where a free-form caption must be localized to a single timestamp, and **action spotting**, where a class label (e.g., "Goal") must be matched to all its occurrences. Both tasks are cast as *cross-video* problems: the model searches across 471 games ($\sim$1.1M sliding windows) with no oracle knowledge of which game is relevant.

We propose a unified **sliding-window retrieval pipeline** that applies to both tasks without architectural changes. Per-frame embeddings are extracted offline at 2 fps; at query time, frames within a $\pm$5s window are mean-pooled and ranked by cosine similarity to the text query. We instantiate this pipeline with two backbones: **CLIP ViT-L/14** [6] (768-dim) and **Qwen3-VL-Embedding-8B** [5] (4096-dim), and complement them with a pairwise fine-tuning recipe for CLIP that requires only a single V100.

To interpret results, we introduce an **oracle diagnostic framework** with three baselines—Oracle(Time), Oracle(Order), and Oracle(Time+Order)—that isolate game-selection error from within-game ranking error. Zero-shot CLIP reaches 0.77% R@1 cross-video but 4.36% with the correct game given, revealing that the dominant bottleneck is game retrieval, not event localization.

Contributions:

- A unified sliding-window pipeline for moment retrieval and action spotting with any VLM encoder.
- An oracle diagnostic framework that decomposes cross-video retrieval error into game-selection vs. within-game components.
- Zero-shot and fine-tuned baselines with CLIP and Qwen3-VL-Embedding on SoccerNet Dense Captioning and SoccerNet-v2.

2 Related Work

Action Spotting. SoccerNet [3] introduced the action spotting task over broadcast soccer with 3 classes; SoccerNet-v2 [2] expanded this to 17 classes across 500 games. Subsequent work trained task-specific temporal models—convolutional spotters, transformer-based E2E-Spot [4]—directly on these labels. Our work differs: we ask whether a general-purpose vision-language model can spot actions zero-shot via text-to-video cosine similarity, without any task-specific training.

Video Moment Retrieval. Natural language video grounding [1] localizes a text query within a *single* video. Cross-video retrieval—finding the right moment across a large corpus—is harder and less studied. The MAD dataset [7] introduced this setting at movie scale. SoccerNet Dense Video Captioning [8] provides the soccer equivalent with 36,894 annotated moments across 471 games. We use this benchmark and evaluate in the cross-video setting.

Vision-Language Models. CLIP [6] learns aligned image and text embeddings via contrastive pretraining on 400M web image-text pairs. Its ViT-L/14 variant produces 768-dim embeddings and runs efficiently on a single V100. Qwen3-VL-Embedding-8B [5] is an 8B-parameter LLM fine-tuned for dense retrieval; it pools the final hidden state at the EOS token to produce 4096-dim embeddings and achieves state-of-the-art results on the MMEB-V2 multimodal retrieval benchmark. Both models share a common interface—encode image and text separately, compare by cosine similarity—which makes them drop-in replacements in our pipeline.

3 Method

3.1 Sliding-Window Retrieval Pipeline

All embeddings are extracted offline at 2 fps and stored as float16 arrays. At query time, we slide a window of width w seconds across the frame sequence. For each center frame t, we mean-pool all frame embeddings within $[t - w/2,\ t + w/2]$ and L2-normalize the result to obtain a window embedding $\mathbf{v}_t$. The text query is encoded once to produce $\mathbf{q}$. Every window is then scored by cosine similarity $s_t = \mathbf{q}^\top \mathbf{v}_t$. For moment retrieval the top-K timestamps are returned; for action spotting the full score sequence is used as the confidence signal for AP computation.

3.2 Feature Extraction

CLIP ViT-L/14. Frames are decoded at 2 fps, resized, and passed through the CLIP vision encoder [6] to produce 768-dim L2-normalized embeddings. Each video half yields a $[T \times 768]$ float16 array ($T \approx 5{,}600$ for a 45-min half). Extraction takes $\sim$85 s per half on a V100.

Qwen3-VL-Embedding-8B. The same 2 fps frames are passed through Qwen3-VL-Embedding-8B [5], loaded via `AutoModelForImageTextToText`. We pool the final hidden state at the EOS token position (`outputs.hidden_states[-1][:, -1, :]`) and L2-normalize to obtain 4096-dim embeddings. Each half yields a $[T \times 4096]$ float16 array. Extraction requires an A100 and takes $\sim$10–15 min per half.

3.3 Oracle Diagnostic Framework

To separate the two sources of cross-video error we define three oracle baselines:

- **Oracle(Time):** the correct game is given; windows within it are ranked *randomly*. This measures the retrieval rate attributable to game selection alone.
- **Oracle(Order):** the correct game is given; windows are ranked by VLM cosine similarity. This isolates within-game visual discrimination.
- **Oracle(Time+Order):** the exact ground-truth timestamp is returned. Trivially 100%; used as a sanity check on evaluation correctness.

The gap Oracle(Time) $\rightarrow$ Oracle(Order) reflects how much visual signal the encoder carries for a given task. The gap Oracle(Order) $\rightarrow$ cross-video zero-shot reflects how much error is introduced by having to retrieve the correct game.

3.4 CLIP Fine-Tuning

We explore two fine-tuning approaches, both training for one epoch on 21,390 anonymized captions from the 281 training games.

Pairwise MSE. For each caption we sample one positive clip (5s window, 4 frames, mean-pooled) and one random frame from a different game as the negative. The loss is:

$$\mathcal{L} = (1 - \cos(\mathbf{q}, \mathbf{v}^+))^2 + \cos(\mathbf{q}, \mathbf{v}^-)^2 \qquad (1)$$

which pushes positive similarity toward 1 and negative toward 0. We use AdamW with LR 10^{-6}, batch size 4, fp16, fine-tuning both encoders on a single 16 GB V100.

InfoNCE. We also train with a symmetric InfoNCE loss using all in-batch pairs as negatives (batch 16 $\rightarrow$ 15 negatives per positive per step). This uses the same positive construction but replaces the MSE objective with cross-entropy over the in-batch similarity matrix, matching the original CLIP training objective. Training requires an A100 due to the larger batch.

4 Experiments

4.1 Datasets

SoccerNet Dense Video Captioning [8] provides 36,894 annotated moments across 471 games from six European leagues. Each annotation contains a free-form description and an anonymized version (player and team names replaced). We use the official split—281 train / 92 val / 98 test games—via `getListGames` from the SoccerNet Python package, and retain only annotations where the event is visible on screen. Annotations from extra time (half = 3) are excluded as no features are extracted for those periods, leaving 36,430 evaluation queries.

Table 1: Moment retrieval at $\delta_t = 5$ s. †98-game test index. ‡471-game full cross-video index.

Method	Backbone	R@1	R@5	R@10
Random	—	0.00%	0.00%	0.00%
Oracle(Time)	—	1.38%	6.76%	13.20%
Zero-shot†	CLIP	1.44%	4.09%	6.00%
Zero-shot‡	CLIP	0.77%	2.23%	3.28%
Zero-shot†	Qwen	0.76%	2.03%	3.12%
Fine-tuned (InfoNCE)†	CLIP	1.42%	4.09%	5.99%
Fine-tuned (pairwise)‡	CLIP	0.41%	1.25%	1.75%
Oracle(Order)	CLIP	4.36%	13.40%	19.39%
Oracle(Order)	Qwen	3.05%	9.36%	14.84%
Oracle(Time+Order)	—	100%	100%	100%

SoccerNet-v2 [2] covers 500 games annotated with 17 action classes: Goal, Corner, Foul, Yellow card, Red card, Yellow→red card, Substitution, Offside, Shots on/off target, Direct/Indirect free-kick, Kick-off, Clearance, Ball out of play, Throw-in, and Penalty. We follow the official evaluation protocol and use all annotations regardless of visibility.

4.2 Implementation Details

All features are extracted at 2 fps with a sliding window of $w = 10$ s and stride 2 s, yielding 1,139,746 windows across the 500 spotting games. Tolerances are $\delta_t \in \{1, 2, 5\}$ s. Features are stored as float16 `.npz` files co-located with the video files on the KAUST IBEX cluster. CLIP runs on V100 (16 GB); Qwen on A100 (40 GB). Both fine-tuning variants use AdamW with fp16 for one epoch; pairwise MSE uses LR 10^{-6}, batch 4; InfoNCE uses LR 10^{-5}, batch 16.

4.3 Evaluation Metrics

Moment retrieval is evaluated with Recall@K for $K \in \{1, 5, 10\}$ at tolerance δ_t. A prediction at rank k is a hit if it falls within δ_t seconds of the single ground-truth timestamp, in the correct game and half.

Action spotting is evaluated with Average Precision (AP) per class and mean AP (mAP) across all 17 classes at $\delta_t \in \{1, 2, 5\}$ s. Predictions are ranked globally by confidence score; a prediction is a true positive if it lies within δ_t of an unmatched ground-truth event in the same game and half.

5 Results

5.1 Moment Retrieval

Results are shown in Table 1. On the 471-game full index, zero-shot CLIP achieves 0.77% R@1—above random but below Oracle(Time) at 1.38%, meaning that knowing the correct game and guessing randomly already outperforms unconstrained CLIP. Oracle(Order) at 4.36% confirms real visual signal within the correct game; the 5.7× gap is almost entirely game-selection failure.

Neither fine-tuning approach improved retrieval. InfoNCE (batch 16) on the 98-game test index yields 1.42% vs. 1.44% zero-shot—statistically indistinguishable. Pairwise MSE on the 471-game index yields 0.41%, a clear degradation from zero-shot 0.77%, suggesting that a single random negative per step does not provide useful gradient signal for cross-video ranking. Zero-shot Qwen (0.76%) is comparable to CLIP despite 20× more parameters; both use the 98-game test index for this comparison.

Table 2: Action spotting mAP (%), 500 games. †Qwen evaluated on 100 test games only.

Method	BB	@1s	@2s	@5s
Oracle(Time)	—	0.60%	1.11%	2.81%
Oracle(Order)	CLIP	1.48%	2.10%	4.64%
Zero-shot	CLIP	0.72%	1.16%	1.83%
Zero-shot†	Qwen	0.75%	1.21%	1.77%

5.2 Action Spotting

Action spotting results are in Table 2 and the per-class breakdown in Table 3. Zero-shot CLIP achieves 1.83% mAP@5s, below Oracle(Order) at 4.64%—the same game-selection gap seen in moment retrieval. The per-class results reveal a clear pattern: visually distinctive and frequent events (Substitution 4.91%, Ball out of play 4.12%, Throw-in 3.80%) score highest, while rare events with few ground-truth instances score near zero (Penalty 0.04%, Yellow→red 0.04%, Red card 0.06%). This is partly a frequency effect—rare classes have few TPs available so precision stays low even with a reasonable score—and partly a domain gap: CLIP was not trained on broadcast soccer and does not reliably associate class labels like "Penalty" or "Offside" with their visual appearance.

6 Discussion

Two failure modes. The oracle framework separates cross-video error into two distinct failure modes. The first is *game-selection failure*: CLIP scores the correct game's windows too low relative to the 470 other games. This is the dominant effect—zero-shot CLIP goes from 0.77% R@1 to 4.36% simply by being told the correct game. The second is *within-game discrimination failure*: even with the correct game, CLIP struggles to precisely localize rare or visually subtle events. Fine-tuning on soccer-specific caption–video pairs directly targets the first mode

Table 3: Per-class AP@5s for CLIP zero-shot cross-video action spotting (500 games, 1,139,746 windows).

Action Class	AP@5s	Action Class	AP@5s
Substitution	4.91%	Indirect free-kick	1.45%
Ball out of play	4.12%	Corner	1.30%
Throw-in	3.80%	Shots on target	0.83%
Direct free-kick	3.50%	Shots off target	0.81%
Yellow card	3.41%	Goal	0.77%
Foul	2.09%	Offside	0.39%
Clearance	1.84%	Red card	0.06%
Kick-off	1.70%	Penalty	0.04%
		Yellow→red card	0.04%
		mAP	**1.83%**

by pulling annotated moments closer to their text descriptions in embedding space.

CLIP vs. Qwen. Qwen3-VL-Embedding-8B (8B parameters, purpose-built for retrieval) performs surprisingly similarly to CLIP ($\sim$400M parameters, general contrastive pretraining): 0.76% vs. 0.77% R@1 on moment retrieval, and 1.77% vs. 1.83% mAP@5s on action spotting. The larger model does not clearly win zero-shot, suggesting the soccer domain gap affects both equally and that scale alone does not compensate for domain mismatch. Qwen does show a meaningful advantage within the correct game (Oracle(Order): 3.05% vs. 4.36% R@1), so its richer representations may be more useful in a reranking setting than in global cross-video retrieval. These gains come at 5× larger embeddings and $\sim$10× slower extraction, requiring an A100 vs. a V100.

Why fine-tuning did not help. InfoNCE with 15 in-batch negatives per step preserves performance (1.42% vs. 1.44% zero-shot) but does not improve it—the in-batch negatives are drawn uniformly from 281 games and are therefore easy (semantically dissimilar) negatives unlikely to reshape the game-selection boundary. Pairwise MSE with a single random negative is even weaker and actively hurts (0.41%), likely because the random frame provides almost no informative contrast for the text query. Both results point to the same root cause: without hard negatives (frames from the same league, same team, or similar action type), the loss provides insufficient gradient signal for cross-video discrimination.

Limitations. Each window is scored independently—there is no temporal reasoning across windows, and mean pooling discards fine-grained structure within the window. The two fine-tuning variants were also evaluated on different index sizes (98 vs. 471 games), making their absolute numbers not directly comparable; a unified evaluation on the full 471-game index is needed.

Future directions. The most promising near-term extension is a two-stage pipeline: use CLIP to retrieve a shortlist of top-50 candidate windows globally, then rerank with Qwen3-VL-Reranker-8B. This combines CLIP's speed with Qwen's stronger semantic understanding while keeping cross-video inference tractable.

7 Conclusion

We presented a unified sliding-window retrieval pipeline for soccer video moment retrieval and action spotting, instantiated with CLIP ViT-L/14 and Qwen3-VL-Embedding-8B. An oracle diagnostic framework decomposes cross-video retrieval error into game-selection and within-game components, revealing that zero-shot CLIP's primary bottleneck is finding the right game (0.77% cross-video vs. 4.36% with the correct game given). Zero-shot CLIP and Qwen perform similarly despite a 20× parameter gap, suggesting domain mismatch dominates over model scale. Two fine-tuning experiments—InfoNCE with 15 in-batch negatives and pairwise MSE with one random negative—neither improved retrieval, pointing to hard negative mining as the critical missing ingredient. The oracle gap—5.7× from cross-video to within-game—provides a clear and quantified target for future retrieval improvements.

References

[1] L. Anne Hendricks, O. Wang, E. Shechtman, J. Sivic, T. Darrell, and B. Russell. Localizing moments in video with natural language. In *Proc. ICCV*, 2017.

[2] A. Deliège, A. Cioppa, S. Giancola, M. Seikavandi, J. Dueholm, K. Nasrollahi, B. Ghanem, T. Moeslund, and M. Van Droogenbroeck. SoccerNet-v2: A dataset and benchmarks for holistic understanding of broadcast soccer videos. In *Proc. CVPR Workshops*, 2021.

[3] S. Giancola, M. Amine, T. Dghaily, and B. Ghanem. SoccerNet: A scalable dataset for action spotting in soccer videos. In *Proc. CVPR Workshops*, 2018.

[4] J. Hong, M. Fisher, M. Gharbi, and K. Fatahalian. Spotting temporally precise, fine-grained events in video. In *Proc. ECCV*, 2022.

[5] Qwen Team. Qwen3-VL: Versatile vision-language models. *arXiv:2601.04720*, 2026.

[6] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In *Proc. ICML*, 2021.

[7] M. Soldan, A. Pardo, J. Alcázar, F. Heilbron, C. Zhao, S. Giancola, and B. Ghanem. MAD: A scalable dataset for language grounding in videos from movie audio descriptions. In *Proc. CVPR*, 2022.

[8] M. Zhou, S. Giancola, A. Cioppa, T. Theiner, N. Somrak, J. Held, and B. Ghanem. SoccerNet dense video captioning. In *Proc. CVPR Workshops*, 2024.